Forecasting Short Horizon Headcount
MMA3001 Project 33880123

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1 - Summary

In HVAC operation throughout office settings, forecasting the occupancy at some short time in the future could enable a more efficient energy utilisation. The system can begin to alter temperatures in rooms prior to occupants entering to ensure comfortable temperatures.

Several numerical methods have been used to predict future occupancy based on current and past information. A base case of 'same as now' was compared to a decision tree and spline fit, with the decision tree producing the lowest RMSE relative to the same as now model on the test set. However, there are limitations to this methodology. Namely, the decision tree has only been trained on data from 5 rooms in 1 building. As such, transferability is untested.

2 - Engineering Problem

HVAC systems are a significant energy expense for businesses. Reaction to changes in the environment is slow, with a lag between turning on and comfortable temperatures being reached. Using numerical methods, the forecasted headcount can be calculated, to begin the lag process earlier. The sensor data provides a current headcount of all 5 rooms in this specific office, however there is no prediction incorporated into the system currently. For a solution to be viable, it must provide a better forecast of occupancy for some time horizon that is equal to the lag period to reach comfortable temperature, as seen in testing results.

The intended use (Figure 1) involves the sampling of occupancy frequently, with the numerical method solution being called on in 5-minute intervals, balancing the need for updated data and system operation frequency. The numerical method would respond to recorded sensor information to inform a central system in control of the HVAC operation. This solution has only been trained and verified on the 5 rooms in the data set detailed in this report, utilising time-stamped sensor data. Applying the same methodology and parameters to a different scenario would not necessarily yield the best results. As such, a clear initial limitation of this solution is the lack of global transferability.

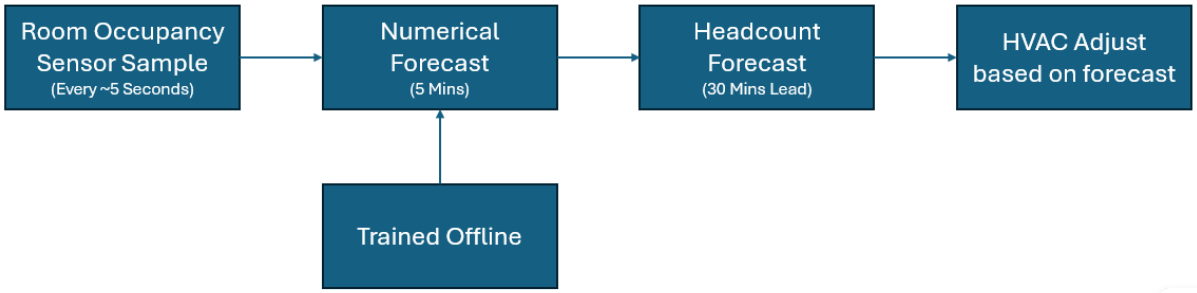


Figure 1: Flowchart of System Operation

3 - Inputs, Outputs, and Data

All data relates to the occupancy in the offices, informing a forecast as mentioned.

Inputs

The input data is the occupancy statistics, recorded by office sensors. The three key inputs are as follows:

	What Is it?	Unit	Allowed Values
Floorspaceid	Which room	Text	Any room name / number
Headcount	Occupancy Count	Number	Zero or more
Collecteddate	Time of sensor recording	Date and time with UTC Offset	Any date and time

Table 1: Inputs Overview

Other data is recorded, relating to sensor numbers and occupancy status, amongst others. However, these are irrelevant for the purposes of this forecast, and as such were ignored.

Outputs

The numerical method outputs a forecast for each room, and other outputs to make it easier for operators to understand the system. See below:

	Meaning	**Unit**	**Intended Use**
floorspaceid	**Which Room**	**Text**	**Identifies which room is to be altered by the HVAC System**
Forecast_for	**Start of period prediction is for**	**Date & Time with UTC Offset**	**Identifying when forecast is for**
Predicted_headcount	**Predicted Occupancy**	**Number**	**Informs HVAC System**
Current_Headcount	**Latest sensor data**	**Number**	**Enables comparison**

Status	Trust / Not Trust	Text	Determines if enough data is present to rely on prediction
Training Range	T or F	Flag	Warns if input is unlike training data, identifying potential variance from standard

Table 2: Outputs Overview

Data

Unusable or invalid data is handled by the system. Rows with missing headcounts are flagged, counted, and dropped from training data. Rejection of clearly incorrect data (missing column for example) is accompanied with an error message to identify recording faults. No raw readings were excluded from the audit. If there is insufficient history to make informed predictions, this is flagged but not treated as an error.

An initial data audit was conducted prior to analysing any numbers. The initial audit of the 9,091,732 readings found the incompleteness of environmental sensors. This data was ignored as accurate predictions couldn't be made. The audit also found extended periods of zero occupancy, during overnight or holiday periods especially. These were accounted for in the functioning, detailed below.

Sensor readings were averaged into 5-minute bins, compromising the computational cost in training and detail. Gaps of up to 15 minutes were filled by interpolating from data either side of the gap. This was the case in 59,995 bins. For gaps longer than 15 minutes up to 1 day, the room was labelled "Assumed empty" if the previous bin had an occupancy of no more than 0.5. For empty periods or other gaps longer than 1 day the data was excluded from training and validation.

4 - Computational Approach

After the cleaning process, the data was split into training, validation and test sets. This was done to ensure the model performs on unseen data, rather than just learning the data. A decision tree was then used to provide a predicted head count at the forecast horizon, a process repeated every 5 minutes. The same feature code is used on training, validation, and live forecasts to ensure the model has the same type of inputs for prediction.

From the inputs as detailed in Section 3, the model extracts the current headcount of the room, previous headcounts at 15-minute intervals prior to the current time, recent

averages, and times. Only the occupancy sensors were utilised in this model, with the time features enabling the model to learn daily or weekly routines.

Decision Tree

The primary model chosen was a decision tree. A decision tree poses several binary questions about features to sort each instance into a group. The depth and leaf size of the decision tree are the variable parts of the function, that were tuned to minimise error on the validation set. The average of the groups are then used as the forecast given previous data. This is a logical solution to this problem, as headcounts change in steps, like a decision tree. As such, posing a series of binary questions more accurately represents the physical restrictions on the model. Additionally, the routine based nature of occupancy and head count is well suited to a decision tree. A unique decision tree was constructed and trained for 15, 30 and 60 minute horizons, as each horizon is a different scenario.

The decision tree has limitations, with only values it has been trained on being usable. Any required extrapolation to previously unseen headcounts must be flagged and the prediction must be used cautiously.

The decision tree was compared against other methods, most notably a simple “Same as now” model and a spline-based model. These will be analysed throughout Sections 5 & 6.

The decision tree depth is how many binary questions are asked by the tree to sort the data as a maximum, with the leaf size being the minimum number of data points in each final group. If a tree has too many leaves or is too deep, it will overfit the data and memorise it. A tree that is too simple will result in the overall trends not being captured. Various depths and leaf sizes were trialled by an algorithm on the validation data, with the best compromise of accuracy and simplicity, meaning an accurate but not overfit, being selected as the chosen model. Ultimately, the 30-minute horizon was assigned a depth of 8, with a leaf size of 200. This model was then re-trained on all training and validation data to expose it to as many scenarios as possible. The test set was not used for this.

5 - Validation Method and Results

Methodology

The raw data was split by recency into 3 groups – Training (60%), Validation (20%), and Testing (20%). It was essential that the testing set was kept isolated, as otherwise the models could learn the test values and provide an inaccurate representation of accuracy. The final decision tree as detailed above was tested on this unseen data, and

compared to baseline and alternative methods (Section 6). The model with the lowest RMSE is deemed to be the best prediction of future headcount.

Results

The results are as follows:

horizon	room	rmse_persistence	rmse_linear_regression	rmse_spline_hybrid	rmse_decision_tree	tree_vs_persistence_%
15	0f3f37f8	0.85	0.76	0.78	0.73	14.04
15	227f1068	0.72	0.64	0.67	0.63	11.76
15	3e5a14df	1.12	1.04	1.07	1.04	6.72
15	941f0352	0.84	0.75	0.77	0.74	11.45
15	cca1ac3e	2.08	2.11	1.98	2.07	0.57
30	0f3f37f8	0.94	0.79	0.82	0.78	16.24
30	227f1068	0.79	0.67	0.71	0.67	15.26
30	3e5a14df	1.33	1.21	1.25	1.21	8.81
30	941f0352	0.94	0.82	0.82	0.79	16.04
30	cca1ac3e	2.92	2.83	2.66	2.73	6.53
60	0f3f37f8	1.00	0.83	0.84	0.82	17.91
60	227f1068	0.85	0.70	0.76	0.71	16.69
60	3e5a14df	1.58	1.40	1.45	1.38	12.91
60	941f0352	1.03	0.88	0.85	0.83	19.97
60	cca1ac3e	3.84	3.45	3.31	3.32	13.55

Table 3: Final Test Results

As such, the decision tree produces a lower RMSE than baseline and hybrid spline fitting by varying margins. The difference is greatest in 60-minute horizons compared to the persistence model, outperforming by up to 20%, which is to be expected given the operation of both systems, verifying expected function. However, the difference in room cca1ac3e is marginal and is worse at 15 minutes when compared to the spline hybrid model by ~5%. Once the cost of implementation and operation is considered, it isn't necessarily true that the decision tree model is the optimal selection.

6 - Comparison and Optimisation

Comparison

A numerical method is only useful to apply if it outperforms the baseline alternative. The most simplistic of these baseline methods is the 'Same as Now' occupancy, where the forecast at all horizons is the same as the current head count. Other alternatives were used, with a spline hybrid model being fit to the data to provide a smooth daily curve for comparison. The list of comparison models is as follows:

Method	What it Does
Same as Now (aka Persistence)	Head Count stays the same
Same Time Last Week	Uses head count from same time last week

Linear Regression	One Straight line formula using same features
Spline Hybrid	Smooth daily curve

Table 4: List of comparison methods

The simple methods, being the persistence and the same time last week, set a baseline of no action taken by the system. If the decision tree or selected model generally cannot outperform this, then it is offering no improvement over assuming the room remains as it is. The linear regression model determines whether any flexibility is required by the decision tree. Table 3 suggests that the improvement of the tree compared to this regression is slight. As such, the added complexity of the tree provides little improvement compared to the regression model.

As Table 3 shows, the decision tree has a smaller RMSE than the persistence method in all tested scenarios. However, in some instances the spline hybrid more accurately predicts head count. Overall, the improvement offered by the decision tree against simple models is substantial. However, when compared to similar but simpler models, the benefits of the decision tree diminish, with most of the benefit of the model captured with the simpler solutions.

Optimisation:

Optimisation was the most time-consuming aspect of the project, due to the large number of possibilities. The parameters opted for was a maximum of 8 depths and 4 leaf sizes, meaning 32 possible trees had to be trained and validated to determine the best selection. Each tree fit and validation took approximately 1.6s initially, meaning prior to optimisation the process exceeded 50 seconds.

Improvement:

To reduce this timeframe, the training set was reduced to every 4th row, meaning the training data was 25% of the initial set. Additionally, a coarse to fine search was conducted, where several widely spaced configurations are tried initially, and then trees are fitted close to the best performing option to reduce the total number of trees fit from 32 to 14. When both are implemented, the training time was cut from over 50s to 6.7s as a final time, whilst maintaining the same accuracy. There are limitations to this coarse to fine search, as in other scenarios the most accurate model might sit as an outlier relative to other high accuracy configurations.

Overall, the optimisation process reduced training time by over 8x, while keeping the same RMSE.

7 - Limitations and Lessons Learned

Limitations

There are many benefits to this decision tree solution. That being said, it is not perfect.

1. There is no quantification of energy saving

The system is implemented to save HVAC energy use for offices and, in turn, reduce energy expenses. While the model provides a more accurate solution that the simple persistence method, it is impossible to quantify the actual benefit derived. It is possible that the benefit is minimal and any costs borne in the implementation of the decision tree offset the savings in energy costs.

2. The lack of extrapolation to wider scenarios

The decision tree model was trained on data from 5 rooms in one building. As such, if the same parameters were applied to other rooms it cannot be guaranteed that this will yield a beneficial outcome. One of the drawbacks of the decision tree is its inability to be effectively extrapolated to data it hasn't seen before. As such, it would have to be trained again on any new environment it's placed in.

And perhaps most importantly,

3. The benefit relative to other methods is modest

When compared to persistence, the model reliably outperforms it. However, the regression and spline models are very similar in performance.

Lessons

The most significant lesson learned is that complex models aren't necessarily better. I spent a significant amount of time working on the decision tree, and only managed to outperform simpler models by a small amount. If I were to do this project again, assuming it's in the real world, instead of chasing the best fit and lowest RMSE at all costs, I would attempt to understand what is required by the system. It might be such that using the persistence methodology is adequate, and the gain in forecasting doesn't justify the computational inputs. As such, I would understand exactly the benefit the client wants and find the simplest model that reaches that requirement.

8 - AI Use and Reflection

I utilised AI throughout the construction of this model, Github repository and report. I used Claude exclusively, and prompted it to generate files, provide explanations and guidance. This was done to increase the speed and efficacy of the code generation, as it can write effective code quickly, whereas my experience is limited. I did not prompt

claude to ‘Do the project’ at the start, and then come back 2 hours later to a completely finished Repo. Instead, I worked with claude, specifying the inclusions of scope, targets for the project, deliverables, formatting etc.

My contribution relative to AI

AI was integral in my completion of this project. The code was almost exclusively written by AI, with me detailing what was required at all steps and overseeing the process. While AI was the thing constructing the code, I verified outputs looked reasonable and pushed back if the directions or assumptions made were not what I wanted. For example, there were times where claude had the wrong requirements listed in requirements.txt for the functions and code it wanted to run. This was causing the code to function incorrectly on my system. I noticed this, and prompted Claude to fix it. There were also times where Claude pulled incorrect numbers from files, which I tested and verified before inclusion in this report.

Also, the spline hybrid comparison was my initiative, suggesting the comparison to a smoother model might provide more insight into the decision tree’s behaviour. For this report, I used AI to assist with finding figures and structure, wrote sections independently, and checked them with Claude and adopted wording from it.

If I had my time again knowing what I know now, I would potentially use different AI models to cross reference outputs. In my experience some are better at specific tasks than others. I use claude most prominently as I believe it to be the best, but a second source is always good.

Claude helped me understand the actual functioning of the decision tree more. At the beginning, I thought it just learned the data not trends, meaning any prediction was pointless. By prompting claude to clarify this, I learned this wasn’t strictly the case. Instead it learns patterns and trends based on observed data. Knowing this and its seeming physical fit, I opted to select this for the model.

All AI use has been logged in the Github Repo. In the AI Use Log.